HOW THIS PRACTICE WORKS: 1. Attempt question in workspace * 2. Look at hints ONLY if stuck * 3. Verify in 'Check Your Method'

Q1 EX-M2B-02 | **EASY** | **JEE MAIN 2023 - 10 APR**

Concept: M-ST02B * Equal Time vs Equal Distance Averages

{{ QUESTION_STEM }}:
A person travels distance x at velocity v_1 and then another equal distance x at velocity v_2 in the same direction. Relate the average velocity v to v_1 and v_2.

Options:

{{ H0_WORKSPACE: Work Here * Try Before Looking at Hints }}

Your Attempt: Option [____] **Reason:** ______________________________

STOP * READ ONLY IF STUCK * H1-H3 HINT LADDER

* {{ H1_NOTICE }}: Both journey legs have the same distance x. Their travel times are unequal: t_1 = x / v_1 and t_2 = x / v_2.
* {{ H2_MODEL }}: Average velocity is defined as Total Displacement / Total Time: v_avg = (x + x) / (t_1 + t_2) = 2x / (x/v_1 + x/v_2).
* {{ H3_START }}: Factor x out of denominator: 2x / [x (1/v_1 + 1/v_2)]. Cancel x, find common denominator v_1 v_2: 2 / [(v_1 + v_2) / v_1 v_2] = 2v_1 v_2 / (v_1 + v_2).

Check Full Method & Takeaways in 'Check Your Method' (Next Page) ->

Q2 EX-M2B-04 | **EASY** | **JEE MAIN 2023 - 30 JAN**

Concept: M-ST02B * Equal Time vs Equal Distance Averages

{{ QUESTION_STEM }}:
A vehicle travels 4 km at 3 km/h and then another 4 km at 5 km/h in the same straight line. Find the average speed of the vehicle.

Options:

{{ H0_WORKSPACE: Work Here * Try Before Looking at Hints }}

Your Attempt: Option [____] **Calculation:** ______________________________

STOP * READ ONLY IF STUCK * H1-H3 HINT LADDER

* {{ H1_NOTICE }}: Notice that the two distances are equal (d_1 = 4 km, d_2 = 4 km). Do NOT take the arithmetic mean (3+5)/2 = 4 km/h - it travels longer at 3 km/h!
* {{ H2_MODEL }}: Use the harmonic mean derived in Q1: v_avg = 2v_1 v_2 / (v_1 + v_2).
* {{ H3_START }}: Substitute: 2(3)(5) / (3 + 5) = 30 / 8 = 3.75 km/h.

Check Full Method & Takeaways in 'Check Your Method' (Next Page) ->

THE APPROVED METHOD RHYTHM: QUESTION RECAP --> WHY THIS WORKS --> METHOD --> ANSWER/CHECK --> CONCEPT TO KEEP

SOLUTION: Q1 EX-M2B-02

CORRECT: (b)

{{ QUESTION_RECAP }}: Equal distances x travelled at v_1 and v_2. Relate v_avg to v_1, v_2.

{{ WHY_THIS_WORKS }}:

Average velocity must be built from total displacement divided by total time. Different speeds contribute to the average in proportion to the TIME spent at each speed, not the distance. Because time = distance/speed, the lower speed takes longer and pulls the average speed down below the arithmetic midpoint (v_1 + v_2)/2.

{{ METHOD_EXECUTION }}:

1. Let each leg have distance x. Total distance = x + x = 2x.
2. Time for leg 1: t_1 = x / v_1.
3. Time for leg 2: t_2 = x / v_2.
4. Total time: t_tot = t_1 + t_2 = x/v_1 + x/v_2 = x (1/v_1 + 1/v_2) = x(v_1 + v_2) / (v_1 v_2).
5. Average velocity: v_avg = Total Distance / Total Time

   v_avg = 2x / [x(v_1 + v_2) / (v_1 v_2)] = 2v_1 v_2 / (v_1 + v_2).

{{ ANSWER_AND_CHECK }}: Correct Option is (b) v = 2v_1 v_2 / (v_1 + v_2).

* Dimensional Check: [v_1 v_2] / [v_1 + v_2] = (m/s)^2 / (m/s) = m/s. [VALID]
* Limiting Case Check: If v_1 = v_2 = v_0, v_avg = 2v_0^2 / 2v_0 = v_0. [VALID]

{{ CONCEPT_TO_KEEP }}:

Permanent Rule: Equal-distance legs always produce the HARMONIC MEAN: 2v_1 v_2/(v_1 + v_2).
Equal-time legs produce the ARITHMETIC MEAN: (v_1 + v_2)/2.

SOLUTION: Q2 EX-M2B-04

CORRECT: (b)

{{ QUESTION_RECAP }}: 4 km at 3 km/h and 4 km at 5 km/h. Find average speed.

{{ WHY_THIS_WORKS }}:

The problem features two equal-distance legs (4 km each). The time spent travelling at 3 km/h is 4/3 h (1.33 h), while time spent at 5 km/h is only 4/5 h (0.8 h). Because 1.33 h > 0.8 h, the car spends more time at 3 km/h. The answer must be strictly less than the arithmetic mean of 4.0 km/h.

{{ METHOD_EXECUTION }}:

1. First-Principles Calculation:

   Total distance = 4 km + 4 km = 8 km.

   Total time = (4/3) + (4/5) = (20 + 12) / 15 = 32/15 h.

   v_avg = 8 / (32/15) = 8 x (15/32) = 15 / 4 = 3.75 km/h.
2. Direct Harmonic Mean Shortcut (from Q1):

   v_avg = 2(3)(5) / (3 + 5) = 30 / 8 = 3.75 km/h.

{{ ANSWER_AND_CHECK }}: Correct Option is (b) 3.75 km/h.

* Sanity Check: 3.75 < 4.0 km/h. Lower speed weighted more heavily. [VALID]

{{ CONCEPT_TO_KEEP }}:

When distance is constant, never average speeds directly. Use the harmonic formula v = 2v_1 v_2 / (v_1 + v_2) or return to total distance over total time.